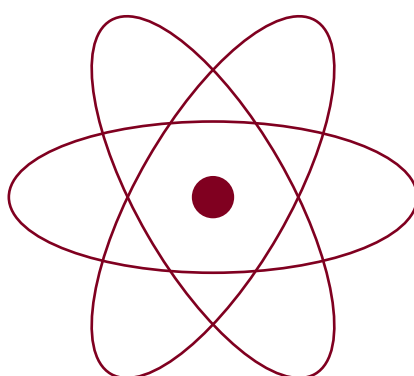


SPhOH

SAUDI PHYSICS OLYMPIAD HUB



Kinematics

Olympiad Problem Set

Olympiad Training • Mechanics

SPhOH Team

FIRST EDITION • 2026

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About This Problem Set

This problem set is part of the training resources of the **Saudi Physics Olympiad Hub (SPhOH)**.

It is designed primarily for students preparing for the Saudi Physics Olympiad and related advanced physics competitions. The purpose of the handout is to provide a focused and substantial collection of problems for deliberate problem-solving practice.

The handout is **problem-centered**. Explanations are intentionally brief and are included only where they help establish the ideas or techniques needed for the problems.

How to Use the Problem Set

Students are encouraged to attempt each problem independently before looking at its hint. A hint should be treated as a way to overcome a specific obstacle, not as a substitute for solving the problem.

For difficult problems, several progressive hints may be provided. Try to use the least amount of assistance necessary before returning to the problem.

Difficulty Levels

The normal problems are divided into three training levels:

- **Level I** — Easy and direct substitution problems.
- **Level II** — Problems requiring combinations of ideas and more substantial reasoning.
- **Level III** — National Olympiad level problems.

Find more at: <https://devfhd.github.io/SPhOH/>.

Kinematics

Topic Note: Kinematic Relations

Kinematics describes motion without considering the forces causing it. Key relationships include:

$$v_{\text{avg}} = \frac{\Delta x}{\Delta t}, \quad a = \frac{\Delta v}{\Delta t}$$

For uniform acceleration a , where v_0 and v represent initial and final velocities while x and Δx represent position and change in position:

$$v = v_0 + at, \quad x = x_0 + v_0 t + \frac{1}{2}at^2, \quad v^2 = v_0^2 + 2a\Delta x$$

In 2D projectile motion, horizontal and vertical motions are independent:

$$x(t) = (v_0 \cos \theta)t, \quad y(t) = (v_0 \sin \theta)t - \frac{1}{2}gt^2$$

The horizontal range R and maximum height H on level ground are given by:

$$R = \frac{v_0^2 \sin(2\theta)}{g}, \quad H = \frac{v_0^2 \sin^2 \theta}{2g}$$

1.1 1D Kinematics and Kinematic Equations

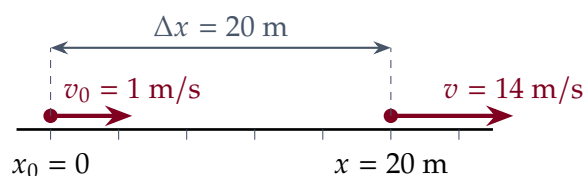
PROBLEM 1

LEVEL I

Acceleration along a Straight Line

Mohammed moves along a straight line with an initial velocity of 1 m/s. His velocity increases uniformly to 14 m/s while covering a distance of 20 m.

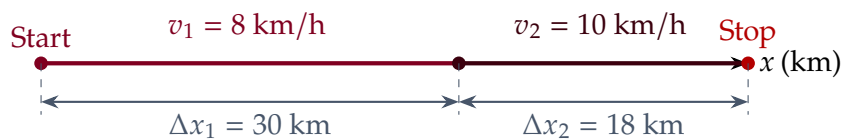
Determine the magnitude and direction of his acceleration.



PROBLEM 2**LEVEL II****Multi-Stage Average Velocity**

Hadi begins moving along a straight line at a constant speed of 8 km/h until he covers a distance of 30 km . Immediately following this, he continues moving along the same line at a constant speed of 10 km/h for a distance of 18 km , after which he comes to a complete stop.

Calculate the average speed for the entire trip (from the start of his motion until he stops).

**PROBLEM 3****LEVEL II****Average Velocity over Split Distance and Time**

A particle covers the first half of its total displacement with a constant velocity v_0 . The second half of the distance is covered in two equal time intervals: moving with a constant velocity v_1 for the first half of that time, and with velocity v_2 for the remaining half of the time.

Calculate the average velocity v_{avg} of the particle averaged over the entire duration of its motion.

PROBLEM 4**LEVEL II****Motorboat and Floating Raft**

A motorboat traveling downstream along a river passes a floating raft at point A. The boat continues downstream for a time interval $\tau = 60 \text{ min}$, then turns around and travels upstream at the same power setting (constant speed relative to the water). The boat meets the raft again at a distance $l = 6.0 \text{ km}$ downstream from point A.

Assuming the river flow velocity remains uniform and constant, determine the speed of the river current.

PROBLEM 5

LEVEL II

2D Motion under Constant Acceleration

A particle begins moving from an initial position vector $\vec{r}_0 = (3\hat{i} - 7\hat{j})$ m with an initial velocity $\vec{v}_0 = 2\hat{i}$ m/s. It moves with a constant acceleration of $\vec{a} = (\hat{i} + 3\hat{j})$ m/s².

- (a) Will the particle cross the x -axis? If so, at what time t and at what x -coordinate?
- (b) Will the particle cross the y -axis? If so, at what time t and at what y -coordinate?

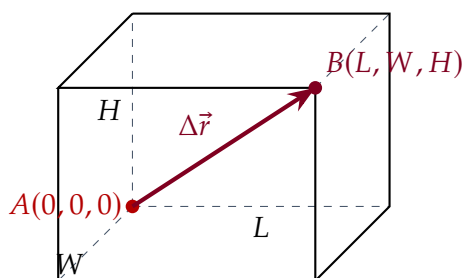
1.2 Vector Displacement in 3D Space**PROBLEM 6**

LEVEL II

3D Box Corner-to-Corner Fly Displacement

A fly flies inside a rectangular room of dimensions $L \times W \times H$ from corner A (at the origin $(0, 0, 0)$) to the diagonally opposite corner B (at (L, W, H)).

Find the magnitude of the fly's displacement vector $\Delta\vec{r}$.

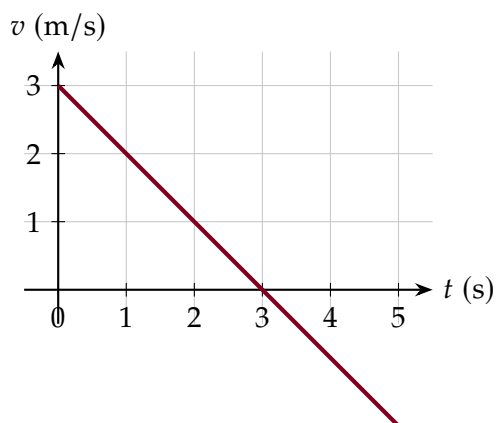
**1.3 Graphical Analysis of Motion****PROBLEM 7**

LEVEL I

Position from Velocity–Time Graph

The graph below represents the velocity $v(t)$ of an object moving along a straight line as a function of time t . It is known that the position of the object at the instant when $v = 0$ is $x = 4$ m.

Determine the initial position of the object.

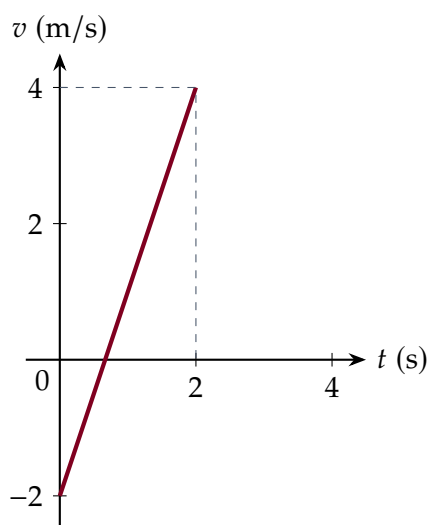


PROBLEM 8

LEVEL I

Acceleration from Velocity–Time Graph

The graph below depicts the velocity $v(t)$ of a particle moving along the x -axis over time. Determine the instantaneous acceleration a of the particle at time $t = 0$.



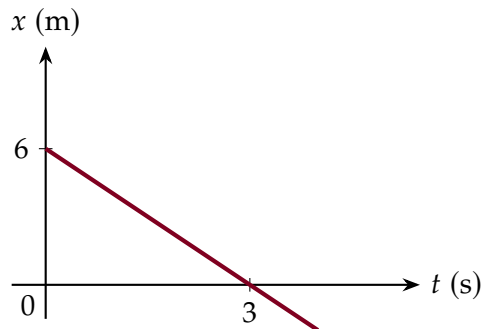
PROBLEM 9

LEVEL I

Instantaneous Velocity from Position–Time Graph

The position $x(t)$ of a particle moving along a straight line is plotted as a function of time t in the figure below.

Determine the instantaneous velocity v of the object at $t = 0$.

**1.4 Free Fall and Vertical Motion under Gravity****PROBLEM 10**

LEVEL III

Continuous Drop of Falling Balls

- (a) A ball is released in air from rest. After time t_0 , it falls down by distance h . How much displacement would it take in the second t_0 interval? (Assume that it did not reach the ground yet)
- (b) Another ball is released from the ground with some initial speed. After time t_0 , its height is h . What is its height After the second t_0 interval? (Assume that its velocity did not reach 0 yet)
- (c) A child standing on the roof of a residential building releases (drops from rest) tennis balls vertically downward at regular time intervals. The rate of dropping is adjusted such that at the exact instant a new ball is released, the previous ball is at some intermediate position in the air, and the ball before that has just struck the ground. At the instant a ball is released by the child, what is the height above the ground of the ball currently in the air? The height of the building is $H = 12$ m.

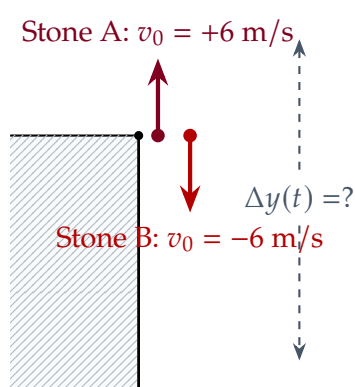
PROBLEM 11

LEVEL II

Separation of Oppositely Thrown Stones

A man standing at the edge of a tall vertical cliff throws two identical stones with the same initial speed $v_0 = 6 \text{ m/s}$: Stone A is thrown vertically upward, and Stone B is thrown vertically downward.

Assuming neither stone has hit the ground below, what is the distance between the two stones after a time $t = 3 \text{ s}$? (Take $g = 9.8 \text{ m/s}^2$)



PROBLEM 12

LEVEL II

Collision of Two Counter-Moving Balls

Two balls are thrown simultaneously along the same vertical line. Ball 1 is launched from the ground vertically upward with an initial speed $v_1 = 20 \text{ m/s}$. Ball 2 is released simultaneously from a height $H = 150 \text{ m}$ above the ground, thrown vertically downward with an initial speed $v_2 = 30 \text{ m/s}$. (Take $g = 10 \text{ m/s}^2$)

- Calculate the time t_{coll} elapsed until the two balls collide, and determine the height y_{coll} above the ground at which the collision occurs.
- Which event occurs first: Ball 1 reaching its maximum height, or the collision between the two balls?
- Determine the distance traveled by the first ball.

1.5 Projectile Motion

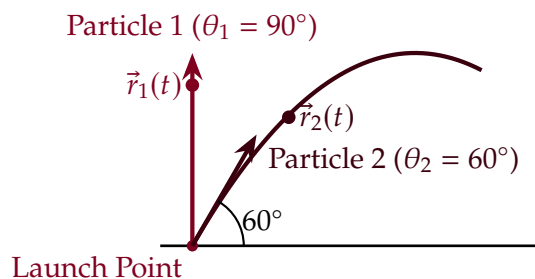
PROBLEM 13

LEVEL II

Diverging Projectiles Distance

Two particles are launched simultaneously from the exact same location with an identical initial speed $v_0 = 25$ m/s. The first particle is projected vertically upward ($\theta_1 = 90^\circ$), while the second particle is launched at an angle $\theta_2 = 60^\circ$ above the horizontal.

Ignoring air resistance, calculate the distance between the two particles at time $t = 1.70$ s after launch.



PROBLEM 14

LEVEL II

Direct Aim at a Vertical Wall

A projectile is launched from ground level at an angle $\theta = 30^\circ$ above the horizontal with an initial speed $v_0 = 40$ m/s, aimed directly at a specific target located on a vertical wall at a horizontal distance $d = 70$ m.

Neglecting air resistance, calculate the vertical deflection from the target (i.e., the vertical distance between the intended target point and the actual impact point on the wall).

PROBLEM 15

LEVEL III

Cliff-Top Projectiles at Opposite Angles

Two identical balls are launched from the edge of a cliff at height $h = 40$ m above level ground with the same initial speed $v_0 = 30$ m/s. Ball 1 is launched at an angle $\theta = 30^\circ$ above the horizontal, while Ball 2 is launched at $\theta = 30^\circ$ below the horizontal.

- (a) Find the difference in time of flight $\Delta t = t_1 - t_2$ between the two balls hitting the ground.
- (b) Find the horizontal distance between their impact points on the ground.
- (c) Compare your answers for time of flight and horizontal distance with those of a projectile launched from level ground with speed $v_0 = 30$ m/s at $\theta = 30^\circ$ above the horizontal. What do you observe?

PROBLEM 16

LEVEL II

Maximum Height to Range Ratio

A projectile is launched from level ground at an angle θ above the horizontal. Let R be its horizontal range and H be its maximum height.

- (a) Derive an expression for the ratio H/R in terms of the launch angle θ .
- (b) If a projectile is fired such that its horizontal range is equal to two times its maximum height ($R = 2H$), find the launch angle θ .

PROBLEM 17

LEVEL II

Fixed Initial Speed with Variable Launch Angle

A cannon fires projectiles at a fixed initial speed v_0 , but the launch angle θ can be varied. The maximum possible horizontal range achievable by the cannon is L .

Determine the horizontal range of a projectile fired at an angle $\theta = 30^\circ$ above the horizontal in terms of L .

PROBLEM 18

LEVEL II

Level-Ground Projectile with Range 60m

A projectile is launched from level ground at an initial angle $\theta = 45^\circ$ and achieves a horizontal range $R = 60$ m. Neglecting air resistance:

- (a) Determine the initial speed v_0 of the projectile.
- (b) Determine the total time of flight T .
- (c) How would the answers to (a) and (b) change if the range remained $R = 60$ m, but the launch angle were increased to $\theta > 45^\circ$? Explain physically.

PROBLEM 19**LEVEL II****Impact Velocity and Vector Direction**

A particle is launched from level ground with an initial velocity $\vec{v}_i$ at an angle θ_i above the positive x -axis. When it returns to ground level, its final velocity vector is $\vec{v}_f$.

Expressed in terms of v_i and θ_i :

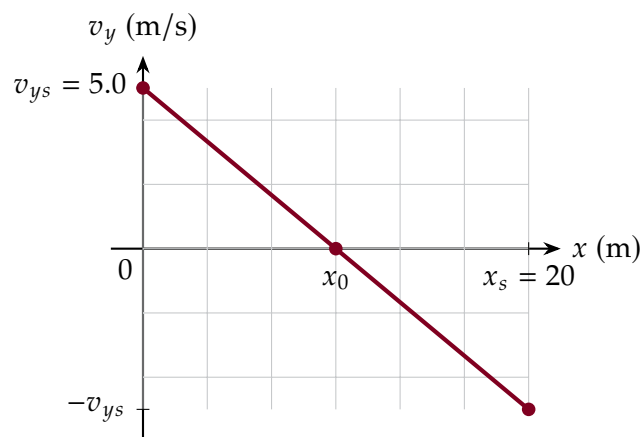
- (a) Find the angle between the initial velocity vector $\vec{v}_i$ and the final velocity vector $\vec{v}_f$.
- (b) Express the sum of velocity vectors $\vec{v}_i + \vec{v}_f$ in unit vector notation $(\hat{i}, \hat{j})$.

PROBLEM 20**LEVEL III****Vertical Component of Impact Velocity against a Wall**

A ball is shot from level ground toward a vertical wall located at a distance x from the launch point. The figure below shows the vertical velocity component v_y of the ball just as it reaches the wall, plotted as a function of the distance x .

The graph scaling is defined by $v_{ys} = 5.0$ m/s and $x_s = 20$ m.

Determine the launch angle θ_0 of the ball.



Hints

The hints below correspond to the problems in the handout. They are provided separately so that students can work through the problem collection without immediately seeing the associated guidance.

@hintlist